VitalCare Wellness Group: Patient Cost and Outcomes Case Study

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Summary

Uninsured patients pay more per visit than insured patients in every one of VitalCare's five regions, and Tamale and Cape Coast are the most expensive places to be treated. I cleaned 1,525 raw visit records down to 1,457 and answered four business questions with Google Sheets pivot tables, charts and an interactive dashboard.

- **Cost:** average cost per visit is GHS 198.73. Tamale (GHS 267.91) and Cape Coast (GHS 228.29) are highest.
- **Seasonality:** typhoid peaks Mar to May, malaria Apr to Jul and respiratory infections Jun to Aug.
- **Outcomes:** insured patients recover most often (88.60%). Diabetes has the lowest recovery rate (72.66%).

All data is synthetic and VitalCare Wellness Group is fictional. This is a portfolio project.

Background and objective

VitalCare Wellness Group is a fictional clinic chain with sites in Accra, Kumasi, Takoradi, Tamale and Cape Coast. The dataset covers patient visits from January 2023 to December 2024. The goal was to find where costs and outcomes differ, so that a management team could plan pricing, staffing and care programmes. Four questions guided the analysis:

1. What is the average cost per visit by region and insurance status?
2. When do the main conditions peak during the year?
3. How do treatment outcomes differ by insurance status?
4. How do treatment outcomes differ by condition?

Data and cleaning

The raw file had 1,525 visit records. After cleaning, 1,457 remain, and every rule below was applied in Google Sheets and logged as it was done.

Step	Issue	Action	Result
1	No untouched original	Duplicated the raw data tab before any edits	Raw Data tab preserved for reference
2	Loose range with no structure	Converted the working range to a table (tblHealthClean)	Filters, structured references and auto-filling formulas
3	Mixed DD/MM and MM/DD dates	Standardized VisitDate to one date format and documented the day/month ordering assumption for ambiguous dates	Consistent dates
4	Duplicate visits	Matched PatientID + VisitDate + Condition and kept the most complete row, then the first occurrence if tied	1,525 to 1,457 rows
5	PatientID reused for different patients	Assigned the correct ID to each patient name	Unique patient IDs
6	Inconsistent Region and Gender labels	Trimmed spaces and normalized casing and abbreviations (for example "Male" instead of "M" or "MALE")	One label per category
7	InsuranceStatus written as Yes/No or Insured/Uninsured	Unified into two consistent values	Insured / Uninsured
8	Invalid ages (for example 187)	Filled from the same patient's other visits, otherwise left blank	32 blanks remain
9	Negative treatment costs	Converted with ABS(), treating them as sign entry errors	No negative costs

Step	Issue	Action	Result
10	Extreme cost outlier (V101230)	Replaced with the median for its condition (GHS 83.92)	No outliers
11	Blank InsuranceStatus	Labeled "Unknown"	36 Unknown rows, no blanks
12	Blank Outcome	Labeled "Not Recorded"	77 Not Recorded rows, no blanks
13	Blank treatment costs	Filled with the median for the same condition	No blanks
14	No month field for seasonality	Added a VisitMonth helper column	Months sort in calendar order

Two decisions are worth explaining. Ages were left blank rather than guessed, because a wrong age is worse than a missing one. Costs were filled with the median for the same condition, not the overall median, because treatment costs differ widely between conditions.

Analysis and findings

Q1. Uninsured patients pay more in every region

Average cost per visit (GHS) across all 1,457 visits is 198.73. Even insured patients in Tamale (243.31) pay more than uninsured patients in Accra (216.00), so Tamale and Cape Coast have a higher base cost regardless of insurance. The percentage gap between uninsured and insured patients is largest in Accra.

Region	Insured	Uninsured	All visits
Tamale	243.31	286.74	267.91
Cape Coast	206.44	258.87	228.29
Takoradi	175.47	202.29	186.75
Kumasi	178.77	193.67	183.68
Accra	165.67	216.00	177.96

Tamale also has 36 visits with an Unknown insurance status (average GHS 254.38), which is why its "All visits" figure is not a simple blend of the two columns. Accra has the most insured visits (387 insured, 125 uninsured), while Tamale has more uninsured than insured (59 insured, 103 uninsured).

Q2. Three conditions are seasonal

Typhoid fever peaks from March to May (34, 26 and 44 monthly visits), malaria from April to July (41, 50, 53 and 52), and respiratory infection from June to August (48, 52 and 34). The other five conditions stay fairly flat through the year, so the seasonal three are where staffing and stock planning matter most.

Q3. Insured patients recover most often

Insurance status	Recovered	Referred to specialist
Insured	88.60%	2.74%
Uninsured	85.85%	4.52%
Unknown	83.33%	2.78%

The gap in recovery between insured and uninsured patients is under three percentage points, while uninsured patients are referred noticeably more often.

Q4. Diabetes has the weakest outcomes

Maternal Checkup has the highest recovery rate (95.22%). Diabetes has the lowest (72.66%), with 10.16% of visits still ongoing and 13.28% referred to a specialist. This fits a long-term condition that is managed over time and not resolved in one visit.

Recommendations

1. **Review pricing and insurance enrolment in Tamale and Cape Coast.** Costs there are highest for insured and uninsured patients alike, and Tamale has the largest share of uninsured and unknown-status patients. Check whether the gap comes from case mix, pricing or supply costs.
2. **Plan staffing and stock around the seasonal peaks.** Prepare for typhoid from March, malaria from April and respiratory infection from June.
3. **Set up a structured follow-up programme for diabetes.** It has the lowest recovery rate and the highest share of ongoing and referred cases.
4. **Improve outcome recording.** "Not Recorded" appears in every group, which weakens any outcome comparison.

These are suggestions from patterns in synthetic data, and each would need to be tested against real clinic data before acting.

Limitations and next steps

- The data is synthetic, so the findings show a method and not real clinic performance.
- The analysis describes patterns only. It does not show why costs or outcomes differ.
- 32 patient ages are blank, so no age-based analysis was done.
- Visit months combine 2023 and 2024, which shows seasonality but hides year-on-year change.
- Next: extend the analysis with the sales and combined datasets, and test cost differences by condition mix within each region.

Tools and skills demonstrated

- **Data cleaning in Google Sheets:** date standardization, duplicate detection with COUNTIFS, median imputation by group, label standardization.
- **Pivot tables:** four analyses on one cleaned table.
- **Dashboard design:** four charts, three slicers (Condition, InsuranceStatus, Region), takeaway lines and a consistent colour palette.
- **Documentation:** a data dictionary, a cleaning log and this case study.